Project Design Phase

Problem – Solution Fit

Date	1 March 2026
Team ID	GACSE-145
Project Name	AI Financial Advisor
Maximum Marks	2 Marks

Problem – Solution Fit:

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why

Purpose:

- ❑ Solve complex problems in a way that fits the state of your customers.
- ❑ Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- ❑ Sharpen your communication and marketing strategy with the right triggers and messaging.
- ❑ Increase touch-points with your company by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly problems.
- ❑ Understand the existing situation in order to improve it for your target group.

Customer Segment

• Students
• Young professionals
• Beginner investors
• Salaried individuals with limited financial knowledge

Problem Statement

Many students and young professionals struggle to manage their personal finances. They find it difficult to track expenses, plan budgets, understand investment options, and make

informed financial decisions. Existing financial tools are often complex and do not provide personalized guidance, which leads to confusion, poor financial planning, and financial stress.

Current Challenges Faced by Users

- Lack of financial literacy and guidance
- Difficulty in tracking expenses and maintaining budgets
- No personalized investment suggestions
- Limited understanding of savings and tax planning
- Overwhelming financial tools that are not beginner friendly

Proposed Solution

The AI Financial Advisor is an intelligent financial assistance system that helps users manage their finances effectively. It provides personalized budgeting assistance, investment recommendations based on risk profile, and AI-powered financial guidance through a chatbot interface.

Key Features of the Solution

1. Personalized Budgeting Assistant – Automatically categorizes expenses and provides alerts.
2. Investment Recommendation System – Suggests suitable investment options based on user risk profile.
3. AI Chat-based Financial Advisor – Allows users to ask financial questions and receive explainable advice.
4. Portfolio Insights – Provides basic analysis of asset allocation and diversification.
5. Financial Education – Simplifies financial concepts for beginners.

Technology Used

- Frontend: Streamlit Web Interface
- Programming Language: Python
- AI Model: Google Gemini API
- Data Processing: Pandas, NumPy
- Visualization: Matplotlib, Seaborn
- Database: SQLite / Local Storage
- Cloud (Optional): Google Cloud / Firebase

Problem–Solution Fit

The proposed AI Financial Advisor directly addresses the financial challenges faced by students and young professionals by providing simple, personalized, and accessible financial guidance. By combining budgeting tools, investment suggestions, and an AI-powered chatbot in a single platform, the solution bridges the gap between complex financial systems and users with limited financial knowledge.

Expected Impact

- Improved financial awareness among young users
- Better budgeting and savings habits
- Increased confidence in making financial decisions
- Easy access to trustworthy financial guidance

1. CUSTOMER SEGMENT(S) (CS)

- Students and young professionals.
- Beginner investors and salaried individuals.
- Users with limited financial literacy seeking automated tools.

6. CUSTOMER CONSTRAINTS (CC)

- **Knowledge:** Lack of deep financial/tax literacy.
- **Affordability:** Professional human advisors are too expensive.
- **Complexity:** Existing tools are not user-friendly for non-experts.

5. AVAILABLE SOLUTIONS (AS)

- Manual tracking (Spreadsheets/Notepad).
- Banking apps (Transactional only).
- Generic YouTube/Finance blogs (Not personalized).

2. JOBS-TO-BE-DONE / PROBLEMS (J&P)

- Manage monthly budget and track expenses effortlessly.
- Plan savings and investments based on risk profiles.
- Understand the "why" behind financial decisions.

9. PROBLEM ROOT CAUSE (RC)

Financial education is disconnected from daily spending tools. Most platforms show what happened (data) but not what to do next (guidance).

7. BEHAVIOUR (BE)

Users currently try to plan savings but often feel overwhelmed. They manually collect receipts (OCR opportunity) and search for finance Q&A on disparate sources.

3. TRIGGERS (TR)

- Unexpected financial stress or overspending.
- Approaching life milestones (buying a car, building an emergency fund).
- Encountering confusing tax or investment jargon.

10. YOUR SOLUTION (SL)

AI Financial Advisor:

- Streamlit UI Dashboard & Chatbot.
- Python logic for saving/debt ratios.
- Gemini API for explainable AI advice.
- Goal-based tracking and investment recommender.

8. CHANNELS OF BEHAVIOUR (CH)

8.1 Online: Web dashboard and Gemini-powered chatbot.

8.2 Offline: Physical receipt handling and daily purchasing decisions at point-of-sale.

4. EMOTIONS: BEFORE / AFTER (EM)

BEFORE		AFTER
Confused, Stressed, Overwhelmed	→	Confident, Secure, Informed